National Stock Exchange of India Limited

Tick by Tick order and trade data via Multicast (MTBT)

A) Master Files for MTBT:

Exchange provides MTBT master files for mapping information about the stream id and token number of the security/contract on specified extranet paths as mentioned below:

Segment	Extranet path	File name
CM	/ftp/common/tbt_masters	cm_contract_stream_info.csv
FO	/faoftp/faocommon/tbt_masters	fo_spd_contract_stream_info.csv fo_contract_stream_info.csv
CD	/cdsftp/cdscommon/tbt_masters	cd_contract_stream_info.csv cd_spd_contract_stream_info.csv
CO	/comtftp/comtcommon/tbt_masters	co_contract_stream_info.csv

The header of above file formats in CM, FO, CD and CO segments are provided in the MTBT API specification document available on the website : https://www.nseindia.com/trade/platform-services-neat-trading-system-protocols .

B) Segment wise details of Multicast IP address and port are as below:

Market participants are advised to monitor and adequately size their infrastructure and systems while subscribing to these feeds.

CM						
Stream ID	Contents	Source 1		Source 2 (Delayed Feed)		Approx. bandwidth utilization
		Multicast IP	Port	Multicast IP	Port	
1	As per master stream file	239.60.60.41	16641	239.60.60.31	10631	8 Mbps
2	As per master stream file	239.60.60.42	16642	239.60.60.32	10632	8 Mbps
3	As per master stream file	239.60.60.43	16643	239.60.60.33	10633	8 Mbps
4	As per master stream file	239.60.60.44	16644	239.60.60.34	10634	8 Mbps
5	As per master stream file	239.60.60.45	16645	239.60.60.35	10635	8 Mbps

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FO						
Stream ID	Contents *	Source 1		Source 2 (Delayed Feed)		Approx. bandwidth utilization
		Multicast IP	Port	Multicast IP	Port	
1	As per master stream file	239.70.70.41	17741	239.70.70.31	10831	20 Mbps
2	As per master stream file	239.70.70.42	17742	239.70.70.32	10832	6 Mbps
3	As per master stream file	239.70.70.43	17743	239.70.70.33	10833	40 Mbps
4	As per master stream file	239.70.70.44	17744	239.70.70.34	10834	18 Mbps
5	As per master stream file	239.70.70.45	17745	239.70.70.35	10835	40 Mbps
6	As per master stream file	239.70.70.46	17746	239.70.70.36	10836	18 Mbps
7	As per master stream file	239.70.70.47	17747	239.70.70.37	10837	40 Mbps
8	As per master stream file	239.70.70.48	17748	239.70.70.38	10838	40 Mbps
9	As per master stream file	239.70.70.49	17749	239.70.70.39	10839	40 Mbps
10	As per master stream file	239.70.70.50	17750	239.70.70.40	10840	40 Mbps
11	As per master stream file	239.70.70.61	17761	239.70.70.71	10871	40 Mbps
12	As per master stream file	239.70.70.62	17762	239.70.70.72	10872	40 Mbps
13	As per master stream file	239.70.70.63	17763	239.70.70.73	10873	40 Mbps
14	As per master stream file	239.70.70.64	17764	239.70.70.74	10874	40 Mbps
15	As per master stream file	239.70.70.65	17765	239.70.70.75	10875	40 Mbps
16	As per master stream file	239.70.70.66	17766	239.70.70.76	10876	40 Mbps
17	As per master stream file	239.70.70.67	17767	239.70.70.77	10877	40 Mbps

18	As per master stream file	239.70.70.68	17768	239.70.70.78	10878	40 Mbps

CD						
Stream ID	**Contents**	**Source 1**		**Source 2 (Delayed Feed)**		**Approx. bandwidth utilization**
		Multicast IP	**Port**	**Multicast IP**	**Port**	
1	As per master stream file	239.80.80.41	18841	239.80.80.31	10831	1.5 Mbps
2	As per master stream file	239.80.80.42	18842	239.80.80.32	10832	1.5 Mbps

CO						
Stream ID	**Contents**	**Source 1**		**Source 2 (Delayed Feed)**		**Approx. bandwidth utilization**
		Multicast IP	**Port**	**Multicast IP**	**Port**	
1	All Commodity Derivatives Contracts	239.90.90.41	19941	239.90.90.51	19951	60Kbps

C) Recovery of tick by tick data

Tick by tick data is available for recovery on TCP. At the Exchange end the recovery requests are managed and serviced through a setup consisting of a load balancer server and multiple recovery servers. If the client end application misses any tick data, it can recover the missed ticks by sending request on TCP to the server IP provided by the Exchange (segment wise IPs and ports provided below). The response messages will be provided on TCP. The TCP connections for recovering messages are not persistent.

A new connection has to be made for each and every recovery request. The recovery requests are served from file stored in recovery server. Due to this factor, there is a lag / delay in the tick data from recovery servers, as compared to direct multicast feed.

Segment	IP Address	Port
CM	172.28.124.34	10984
FO	172.28.124.30	10990
CD	172.28.124.32	10970
CO	172.19.125.84	10960

D) Order Snapshot Recovery for Multicast Tick by Tick broadcast

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MTBT Order Book Snapshot Recovery Server provides latest picture of the order book, constructed from MTBT feed. This Order Book Snapshot is refreshed every 30 seconds. Client requesting this snapshot recovery will get all the outstanding orders in the order book till the last refresh time for a particular stream.

Parameters for connecting for Order Snapshot Recovery		
IP of Solace	172.28.124.44	
Port of Solace	10987	
User Name	member_cm_prod	
Password	LxJRe#0414	
Message VPN Name	od_cm_prod	
Segment	**Stream ID**	**QUEUE NAMES**
CM	1	lvq.nse.cm.od.1.orderbook
	2	lvq.nse.cm.od.2.orderbook
	3	lvq.nse.cm.od.3.orderbook
	4	lvq.nse.cm.od.4.orderbook
	5	lvq.nse.cm.od.5.orderbook

Parameters for connecting for Order Snapshot Recovery		
IP of Solace	172.28.124.40	
Port of Solace	10986	
User Name	member_fo_prod	
Password	YoEvx#9675	
Message VPN Name	od_fo_prod	
Segment	**Stream ID**	**QUEUE NAMES**
FO	1	lvq.nse.fo.od.1.orderbook
	2	lvq.nse.fo.od.2.orderbook
	3	lvq.nse.fo.od.3.orderbook
	4	lvq.nse.fo.od.4.orderbook
	5	lvq.nse.fo.od.5.orderbook
	6	lvq.nse.fo.od.6.orderbook
	7	lvq.nse.fo.od.7.orderbook
	8	lvq.nse.fo.od.8.orderbook
	9	lvq.nse.fo.od.9.orderbook
	10	lvq.nse.fo.od.10.orderbook
	11	lvq.nse.fo.od.11.orderbook
	12	lvq.nse.fo.od.12.orderbook
	13	lvq.nse.fo.od.13.orderbook

	14	lvq.nse.fo.od.14.orderbook
	15	lvq.nse.fo.od.15.orderbook
	16	lvq.nse.fo.od.16.orderbook
	17	lvq.nse.fo.od.17.orderbook
	18	lvq.nse.fo.od.18.orderbook
Parameters for connecting for Order Snapshot Recovery		
IP of Solace	172.28.124.42	
Port of Solace	10988	
User Name	member_cd_prod	
Password	48Qd5d	
Message VPN Name	od_cd_prod	
Segment	**Stream ID**	**QUEUE NAMES**
CD	1	lvq.nse.cd.od.1.orderbook
	2	lvq.nse.cd.od.2.orderbook

Important usage guidelines for connecting to Order Book Snapshot recovery:

Members are suggested to use snapshot recovery whenever they have lost large number of ticks on multicast or they have joined the MTBT feed very late. This will reduce the time to recover all the missed ticks through recovery and build the order book up to the latest point.

At the Exchange end the snapshot recovery is disseminated through Solace Application / Hardware. The snapshot recovery needs to be requested by client applications through Solace API. After receiving the snapshot message, the client applications are supposed to disconnect. It is possible that the client applications can get disconnected during recovery of snapshot message, due to snapshot message being refreshed at the Exchange end. In such cases client applications can again try to recover the snapshot message.

Members are expected to use 7.32.0.3 version of SOLACE API which can be downloaded from https://solace.com/downloads/ for best experience.

General Notes applicable for all types of market data broadcast:

- *The size of broadcast indicated above is an estimated approximation and shall change depending upon the market activity. Thus, Market participants are strongly advised to monitor and adequately size their infrastructure and systems while subscribing to these feeds.*

- *With respect to using Exchange data, Members may kindly take note of Exchange circulars NSE/MSD/64333 dated October 03, 2024, NSE/COMP/56426 dated April 20, 2023 and NSE/MEM/26958 dated June 19, 2014. Exchange data can be used by Member's clients who are registered for trading with the member on the Exchange in any segment.*
- *Members shall be solely responsible for implementing and maintaining redundancy for market data broadcast subscribed from NSEIL. NSEIL shall not be held liable in any manner whatsoever including but not limited to for any disconnection, excess bandwidth utilization, related latency issues' etc. which arises out of the member's decisions on market data broadcast configurations.*
- *National Stock Exchange of India Limited ("NSEIL") strongly recommends its member to build resilience and redundancy in their subscription to market data broadcast provided by NSEIL, to ensure continuity of their business operations. NSEIL has made enabling provisions for members to implement redundancies by providing two sources of data and/or data recovery mechanisms. Members shall be solely responsible for implementing and maintaining redundancy for market data broadcast subscribed from NSEIL. NSEIL shall not be held liable in any manner whatsoever including but not limited to for any disconnection, excess bandwidth utilization, related latency issues' etc. which arises out of the member's decisions on market data broadcast configurations.*

Annexure 3

Members are requested to take note of below advisory and guidance regarding market data broadcast parameters:

A. Advisory for members subscribing to Price volume (5 depth 1 second refresh), other related data, master updates, market open/close status message broadcast etc.

- NSEIL has made enabling provisions for members to implement redundancies by providing two sources of data broadcast parameters namely Source 1 and source 2.
- Exchange has provided option for Members to add the broadcast configuration for both sources in the Exchange provided UDP application for windows as well as linux OS.
- Member can configure the sources 1 & 2 in active – active manner / active – passive manner as per their business requirement.
    - Exchange broadcasts the market data on both the sources/channels simultaneously. Members are required to select one of the option
        - Option1 - active-active manner i.e. both the sources/channels are configured at member's end and market data is received by members on both the sources/channels OR
        - Option2 - active-passive manner i.e. both the sources/channels are configured at member's end but market data is received by members on the source which is activated by member.
    - The Exchange has multiple backup systems and processes besides having various checks and balances to ensure proper functioning of the trading systems. However, the possibility of an event where Exchange is not able to broadcast the market data on one of the source/channel cannot be ruled out. Hence, the above action will help members to quickly shift the source at

their end and mitigate the risk of business disruption in case one of the source/channel goes down.

- It is strongly recommended that the members build resiliency in their implementation by choosing any of the above-mentioned options. Members shall be solely responsible for implementing and maintaining redundancy for market data broadcast subscribed from the Exchange and Exchange will not be liable for any business disruption at member's end arising due to no action by members.

- For members using Exchange provided UDP application, option is available to members to start / stop the broadcast for all / respective streams as per their convenience in the UDP application keeping in mind the bandwidth requirements of the broadcast streams.
- Manuals for UDP application for Windows as well as Linux are available on Extranet for reference.

B. Advisory for members subscribing to Multicast Tick By Tick Feeds (MTBT)

- The Tick Data broadcast feed provided by the Exchange is available on two separate Multicast Channels (Source 1 and Source 2) for each Stream on active-active basis. On one multicast channel the feed is delayed and lagging behind the other feed. Please refer the captioned circulars for details regarding the Source 1 and Source 2 multicast IP and ports.
- Kindly note that the MTBT data broadcast being on UDP, there can be occasional drops in the broadcasts provided by the Exchange due to the same. It is an expected behaviour.
- It is recommended that the end user application should subscribe to both multicast channels for each stream and receives the data from both the channels in active-active manner. This needs to be done to ensure that data loss is minimized for the receiving applications as there is a possibility of data loss in multicast protocol.
- If the end user application misses any tick data from both the active-active channels it can recover the ticks by sending recovery requests to recovery server on TCP. The recovery requests are served from file stored in recovery server. Due to this factor, there is a lag / delay in the tick data from recovery servers, as compared to direct multicast feed.
- Members are requested to refer the API's for MTBT, wherein we have provided the tuning parameter for receiving MTBT as well as guidelines for recovery.
- Exchange also provides facility for snapshot recovery. Members are suggested to use snapshot recovery whenever they have lost large number of ticks on multicast or they have joined the MTBT feed very late. This will reduce the time to recover all the missed ticks through recovery and build the order book up to the latest point.
- API for MTBT can be accessed from NSE website under the following link: https://www.nseindia.com/trade/platform-services-neat-trading-system-protocols

Relevant Circulars:

Download No.	Date
NSE/MSD/54696	December 02, 2022
NSE/MSD/62785	July 04, 2024
NSE/COM/66509	February 05, 2025